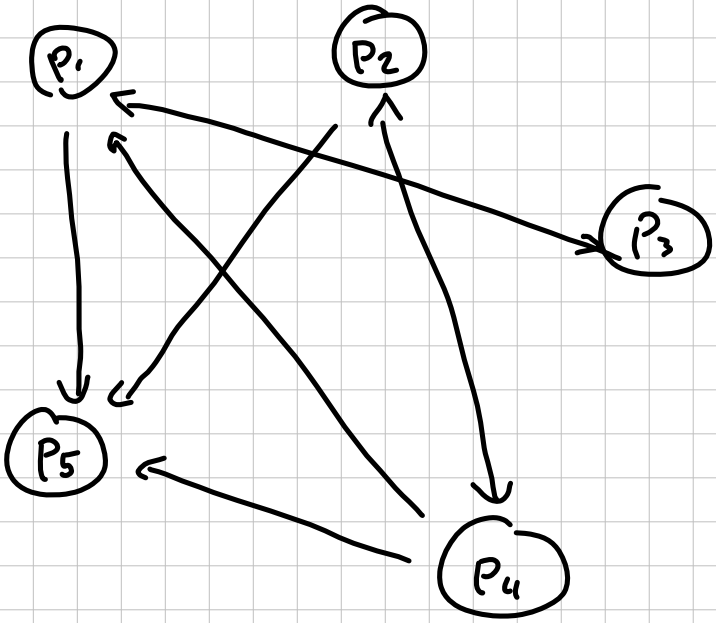
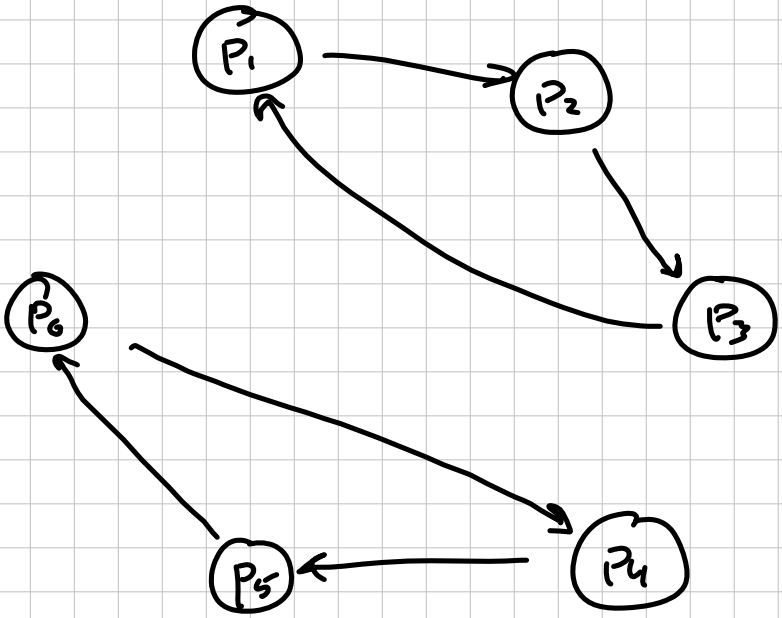
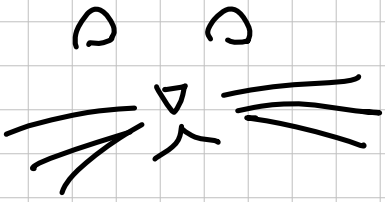
Opgave 1



Opgave 2





Opgave 7

B

A	P_1	P_2	P_3	P_4	P_5
P_1	0	0	1	0	1
P_2	0	0	0	1	1
P_3	1	0	0	0	0
P_4	1	1	0	0	1
P_5	0	0	0	0	0

(directed graph)

$A \longrightarrow B$

direction

Opgave 8

$$
\begin{array}{c|ccccc}
 & P_1 & P_2 & P_3 & P_4 & P_5 \\
\hline
P_1 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\
P_2 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\
P_3 & 1 & 0 & 0 & 0 & 0 \\
P_4 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{1}{3} \\
P_5 & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5}
\end{array}
$$

i ↓ j →

sends the "surfer" to a random node because P_5 terminates

Opgave 7

$$
\begin{array}{c|ccccc}
 & P_1 & P_2 & P_3 & P_4 & P_5 \\
\hline
P_1 & 0 & 0 & 1 & 0 & 1 \\
P_2 & 0 & 0 & 0 & 1 & 1 \\
P_3 & 1 & 0 & 0 & 0 & 0 \\
P_4 & 1 & 1 & 0 & 0 & 1 \\
P_5 & 0 & 0 & 0 & 0 & 0
\end{array}
$$

A

B

matrix

does NOT send the "surfer" anywhere, thus terminating

Opgave 14

rest of chance from $(1-d)$

$$PR_{n+1}(p) = (1-d)\frac{1}{N} + d\sum_{q \in in(p)} \frac{PR_n(q)}{deg^+(q)}$$

x any node

chance of hopping to random page

weight of page pointing to p

all vertecies pointing to p

chance of given $q \to p$ transfer

In short:

new chance of p is found by adding a set random deviation to the old chance of p.
This is done by a completely random choice weighted by d, and ALL the chances of other vertecies pointing to p.

Opgave 16

$$PR_{n+1}(p) = \underbrace{(1-d)\frac{1}{N}} + d\sum_{q \in in(p)} \frac{PR_n(q)}{deg^+(q)}$$

$$x_{n+1} = \frac{1-d}{N} e + d L x_n$$

transform into vector $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$ so we can operate on it with matricies

(multiplication by 1 does fuck all)

here any item is of form $\frac{1}{deg^+(q)}$

thus

Lx_n gives items of form $\frac{PR_n(q)}{deg^+(q)}$

(cause $x_n = \begin{bmatrix} PR_n(q_1) \\ \vdots \\ PR_n(q_n) \end{bmatrix}$

part 1

part 2

Opgave 16 (part 2)

$$x_{n+1} = M_d x_n$$

$$\downarrow$$

$$x_{n+1} = \left(\frac{1-d}{N} E_N + dL\right) x_n$$

$$\downarrow$$

$$x_{n+1} = \frac{1-d}{N} E_N x_n + d L x_n$$

$$\downarrow$$

$$E_N x_n = \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix} x_n, \text{ where } \sum_{i=1}^{N} x_{n_i} = 1$$

$$\Downarrow$$

every row in $E_N x_n = 1$

$$\Downarrow$$

$$E_N x_n = e, \text{ where } e = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \updownarrow N$$

$$\downarrow$$

$$\underline{\underline{x_{n+1} = \frac{1-d}{N} e + dLx_n}}$$